

18.06 Linear Algebra - Lecture 7

Solving $Ax = 0$: Pivot Solution, Special Solutions

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$$

→ Elimination but now we continue even if there are 0 in the pivot positions

$$m \times n = 3 \times 4$$

→ Computing the Nullspace ($Ax = 0$)

→ Pivot Variables - free variables

→ Special Solutions - $\text{rref}(A) = R$

$$A \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

Pivot Columns Free Columns

→ In echelon form
↳ "stair"

→ Rank of A

↳ = # of pivots (A rank = 2)

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 0 \\ 2x_3 + 4x_4 &= 0 \end{aligned}$$

$$x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \rightarrow x = c \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\rightarrow \text{Rank}(A) = 2 \rightarrow 2 \times \text{pivot variables } (r = 2)$$

$$\therefore m - r = 4 - 2 = 2 \text{ free variables}$$

$R \rightarrow$ Reduced Row Echelon Form $\rightarrow \text{rref}$

↳ zeros above + below pivots.

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R = \text{rref}(A)$$

↳ Notice: $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ in pivot rows

↳ Notice: $\begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}$ in free columns

↳ Signs will swap

$$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$$

$\uparrow$ $\uparrow$
 (r) pivot cols $(n-r)$ free cols

$I \rightarrow$ identity
 $F \rightarrow$ Free

→ Nullspace:

$$R_x = 0 \rightarrow RN = 0 \rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix}$$

$$\begin{bmatrix} R_x = 0 \\ I \quad F \end{bmatrix} \begin{bmatrix} x_{\text{pivot}} \\ x_{\text{free}} \end{bmatrix} = 0$$

$$x_{\text{pivot}} = -F \cdot x_{\text{free}}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \quad (m \cdot n = 4 \times 3)$$

↳ $1 + 2 = 3$ Col 3 is dependent on Col 1 + 2.

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U \quad (r = 2 \text{ again!})$$

↳ Free Column: $n - r = 3 - 2 = 1$

$$\rightarrow x = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \dots \begin{bmatrix} -1 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

↳ Start with free column

↳ Give free variable to a convenient value

$$\rightarrow x = C \begin{bmatrix} -1 \\ -1 \end{bmatrix} \quad C \begin{bmatrix} -F \\ I \end{bmatrix}$$

↳ Solve remain x_i values.

↳ Find Nullspace → this is the nullspace.

↳ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$ → just the "basis" of the nullspace

$$\rightarrow N = \begin{bmatrix} -F \\ I \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

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Textbook Notes - Section 3.2

• Section 3.2 Nullspace of $A \rightarrow$ Solving $Ax = 0$ & $Rx = 0$

↳ Nullspace + Column Space

• Nullspace

↳ 1st two of the Four Fundamental Spaces

The nullspace $N(A)$ is the set of all solutions x to $Ax = 0$

- The nullspace is always a subspace
- The zero vector is always in the nullspace
- Any combination of solutions is also a solution

• Pivot Variables vs Free Variables

After eliminating a matrix to echelon form U (or reduced form R):

- Pivot variables - columns that contain a pivot.
 - these are determined by the free variables.
- Free variables - columns with no pivot.
 - can be set to anything
 - they generate the Nullspace
 - Number of free variables = $n - r$

↳ n is # of matrix columns

↳ r is the matrix rank.

• Special Solutions

For each free variable, there is 1 special solution

- (1) Set a free variable to 1.
- (2) Set all other free variables to 0.
- (3) Back-substitute to find the pivot variables.

↳ The Nullspace is all linear combinations of the special solutions.

• Reduced Row Echelon Form (R)

↳ Going from echelon form U to reduced form R , makes special solutions easier to read off directly.

- R has:
- ⊙ 1s in every pivot position ⊙
 - ⊙ 0s above & below every pivot (U is just below) ⊙
 - ⊙ Identity (I) in the pivot columns. ⊙

Free columns of R directly reveal the special solutions

↳ just flip the signs of the free column entries

• Rank:

The rank r of a matrix is the number of pivots it has.

Rank is a measure of the "size" of a matrix beyond its dimensions:

↳ r pivot variables ; $n-r$ free variables

↳ Nullspace contains only $x=0$ when $r=n$ (no free variables)

↳ Column space fills all of $\mathbb{R}^m$ when $r=m$ (no zero rows)

• Wide Matrices

→ If a matrix has more columns than rows, it always has at least 1 free variable.

∴ $Ax=0$ always has nonzero solutions

↳ the Nullspace is never just the zero vector.

• Rank One Matrices

A rank one matrix has the form $A = uv^T$

↳ every row is a multiple of one pivot row.

↳ every column is a multiple of one pivot column.

The Column Space is a single line, the Nullspace is very large.

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Solving $Ax = b$: Row Reduced Form: R

Complete Solution to $Ax = b$

Rank r

$r = m$: solution exists $\cdot r = n$: solution ~~undefined~~ unique.

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= b_1 \\ 2x_1 + 4x_2 + 6x_3 + 8x_4 &= b_2 \\ 3x_1 + 6x_2 + 8x_3 + 10x_4 &= b_3 \end{aligned}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{bmatrix}$$

$\hookrightarrow$ Augmented Matrix $[A \ b]$ $\nearrow$ pivot columns

OK $b = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix} \rightarrow \dots b = \begin{bmatrix} b_1 \\ b_2 - 2b_1 \\ b_3 - b_2 - b_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$

$\hookrightarrow$ Solvability Condition on b :

$Ax = b$ Solvable when b is in $C(A)$ "column space of A "

$\hookrightarrow$ If a ~~comb.~~ comb. of rows of A gives zero row, then same comb. of entries of B must give 0.

To find complete solution to $Ax = b$

① $x_{\text{particular}}$: Set all free variables to zero
Then solve $Ax = b$ for pivot variables.

$$\left. \begin{aligned} x_1 + 2x_3 &= 1 & \therefore x_1 &= -2 \\ 2x_3 &= 3 & x_3 &= 3/2 \end{aligned} \right\} x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

$\neq \downarrow$
 $x_2 = 0$
 $x_4 = 0$

② $x_{\text{Nullspace}}$: $x_N \rightarrow$ any single vector in the Nullspace.

③ Complete: $x_c = x_p + x_N$
 $\hookrightarrow$ any vector from Nullspace.

$$x_c = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$\hookrightarrow x_1 + 2x_2 + 2x_3 + 2x_4 = 0$

Why?

$Ax_p = b$

Sum $Ax_N = 0$

$A(x_p + x_N) = b$

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Textbook Notes — Section 3.3

• The Complete Solution to $Ax = b$

• Core Idea

The complete solution to $Ax = b$ is:

$$x_c = x_p + x_N$$

↳ where x_p is one particular solution

↳ x_N is any vector in the nullspace

→ The particular solution accounts for the right side of b .

→ The nullspace vectors account for all the freedom in the solution.

• Process to Find the Complete Solution

Step 1 → Augment & Eliminate → Form $[A \mid b]$ and row reduce to $[R \mid d]$
↳ Row operations apply to both sides simultaneously

Step 2 → Check Solvability → If any zero row in R has a non-zero entry in d , the system is inconsistent → no solution
↳ Condition: All zero rows of R must correspond to zeros in d .

Step 3 → Find x_p → Set all free variables to zero
↳ Back-substitute to find the pivot variables.
↳ Provides one particular solution.

Step 4 → Find x_N → Find special solutions to $Rx = 0$ (see Section 3.2)
↳ The Nullspace is all combinations of these.

Step 5 → Combine → $x = x_p + x_N$ is the complete solution.

- Why This Works

If $Ax_p = b$ and $Ax_n = 0$, then:

$$A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b$$

The Nullspace vectors do not disturb the particular solution $\rightarrow$ they only shift to other valid solutions.

- Full Column Rank ($r = n$)

When every column has a pivot, there are no free variables.

The Nullspace contains only $x = 0$.

$\hookrightarrow$ Consequences:

- $Ax = b$ has at most one solution — either zero or one, never infinitely many.
- Typical case is tall and thin ($m > n$, overdetermined) — usually no sol exists
- When a solution does exist, it is unique.

- Full Row Rank ($r = m$)

When every row has a pivot, there are no zero rows in R

The solvability condition $0 = 0$ is always satisfied.

$\hookrightarrow$ Consequences:

- $Ax = b$ is solvable for every b .
- Column space is all of $\mathbb{R}^m$
- There are $n - m$ free variables, so infinitely many solutions.
- Typical case is short and wide ($m < n$, underdetermined).

• Four Cases

↳ Everything depends on rank r relative to m and n .

Case	Condition	Solutions
Square Invertible	<u>$r = m = n$</u>	Exactly 1
Short & Wide	<u>$r = m < n$</u>	Infinitely Many
Tall & Thin	<u>$r = n < m$</u>	0 or 1
Rank deficient	<u>$r < m$ & $r < n$</u>	0 or Infinitely Many.

• Overview

↳ Section 3.3 ties together Chapter 2+3.

↳ Elimination does not just solve $Ax = b \rightarrow$ it reveals

↳ the rank

↳ the nullspace

↳ the column space

↳ the complete solution structure

↳ Particular Solution (x_p) $\rightarrow$ exists in row space of A

↳ Nullspace vectors x_N are Orthogonal to it.

↳ Together they provide the full solution.

18.06 Linear Algebra - Lecture 9

Independence, Basis and Dimension

- Linear independence
 - Spanning a space
 - Basis and Dimension
- } Vectors ; not Matrices.

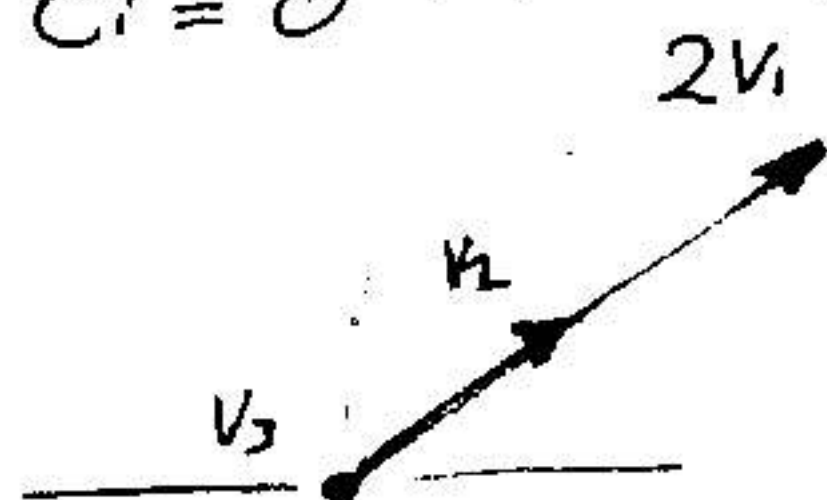
- Suppose A is m by n with $m < n$
Then there are nonzero solutions to $Ax = 0$
(more unknowns than equations)

↳ Reason: There will be free variables.

Independence

Vectors $x_1, x_2, \dots, x_n$ are independent if no combination gives zero vector (except the zero comb,

$$\begin{pmatrix} c_1 x_1 + c_2 x_2 + \dots + c_n x_n \neq 0 \\ \text{all } c_i = 0 \end{pmatrix}$$



$$\begin{cases} 2v_1 - v_2 = 0 \\ 0v_2 + 5v_3 = 0 \end{cases} \text{Ind.}$$

$$\rightarrow A = \begin{bmatrix} 2 & 1 & 2.5 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

- Repeat when $v_1, \dots, v_n$ are columns of A
- They are independent if Nullspace of A is $\{ \text{zero vector} \}$ Rank = n
 ↳ No free variables
- They are dependent if $Ac = 0$ for some nonzero c Rank $< n$
 ↳ Yes free variables

• Spanning a Space

Vectors $V_1, \dots, V_m$ span a space means:

↳ The space consists of all combinations of those vectors.

• Basis

Basis for a ~~vector~~ space is a sequence of vectors $V_1, V_2, \dots, V_d$ with two properties:

- They are independent.
- They span the space.

e.g.:

Space is $\mathbb{R}^3$

One basis is: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Another: $\begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 3 \\ 8 \end{bmatrix}$

$\left. \begin{array}{l} \text{⌚ test by putting in columns of a matrix} \\ \text{⌚ eliminate matrix} \\ \text{⌚ check if pivots in every column.} \end{array} \right\} \text{Test process}$

→ $\mathbb{R}^n$ n vectors give a basis if the $n \times n$ matrix with those columns is invertible.

→ Many basis for a space.

↳ All basis have the same number of vectors $\mathbb{R}^n \Rightarrow n$ vectors.
↳ Dimension

• Dimension

↳ The number of vectors in any basis of any a space.

• Space is $C(A)$ $N(A)$

$$\begin{bmatrix} 1 & 2 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

• • •

$$\begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}$$

$\text{Rank}(A) = \# \text{ pivot columns} = \text{dimension of the column space.}$

↳ Not the dimension of matrix A
↳ dimension of subspace $C(A)$

↳ Not the rank of a subspace
↳ rank of matrix A .

• ↳ Another Basis of A : $\begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 7 \\ 5 \\ 7 \end{bmatrix}$

↳ ~~$\dim(A)$~~ $\dim C(A) = 2 = r$

• ↳ Vectors in the Nullspace

$$N(A) = \begin{bmatrix} -1 \\ -1 \\ 0 \end{bmatrix}; \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \rightarrow \underline{\underline{\dim N(A)}} = \# \text{ free variables} = \underline{\underline{n - r}}$$

18.06 Linear Algebra - Lecture 9

Textbook Notes - Section 3.4

3.4 - Independence, Basis & Dimension:

Four Core Ideas

1. Linear Independence → no redundant vectors
2. Spanning → enough vectors to cover the space
3. Basis → independent vectors that span
(not too many, not too few)
4. Dimension → the number of vectors in any basis

Linear Independence

Matrix definition → the columns of A are linearly independent when the only solution to $Ax = 0$ is $x = 0$.
→ the Nullspace contains only the zero vector.

Vector definition → Vectors $v_1, \dots, v_n$ are linearly independent if:

$$x_1 v_1 + x_2 v_2 + \dots + x_n v_n = 0 \Rightarrow \text{all } x_i = 0$$

↳ the only combination that produces zero is the trivial one (all $x_i = 0$).

↳ if any nonzero combination produces zero, the vectors are dependent - at least one vector is a combination of the others.

Geometric intuition: → Three vectors in the same plane are dependent,
→ Three vectors not in the same plane are independent.

Critical Rule: → Any n vectors in $\mathbb{R}^m$ must be dependent if $n > m$
→ More columns than rows forces free variables, which forces nonzero solutions to $Ax = 0$.

• Spanning a Space

- ↳ A set of vectors spans a space if every vector in the space can be written as a linear combination of them.
- ↳ The columns of A span its column space by definition.
- ↳ They may span it with ^o redundancy (dependent columns) or efficiency (independent columns)

• Basis

- A basis for a vector space is a set of vectors that is both
 - ↳ linearly independent $\rightarrow$ no redundancy
 - ↳ spanning $\rightarrow$ no gaps
- The basis is a "just right" set $\rightarrow$ neither too many nor too few vectors.
- Uniqueness of Representation:
 - ↳ Every vector in the space can be written as a combination of basis vectors in exactly one way.
 - ↳ Follows directly from independence — if there were two ways, their difference would be a nonzero combination equalling zero, ~~contradicting~~ contradicting independence.

→ Examples

The standard basis for $\mathbb{R}^n$ is the columns of the identity matrix I .
But there are infinitely many valid bases.

The columns of any $n \times n$ invertible matrix form a basis for $\mathbb{R}^n$
↳ invertibility guarantees both independence and spanning

→ Finding Bases from a matrix :

- Column space of A → use pivot columns of A (not R)
- Row space of A → use the nonzero rows of R
- Nullspace of A → use the special solutions

Important → the pivot columns of R are a basis for the column space of R , but not necessarily for the column space of A .

→ Column spaces of A and R differ - always take pivot columns from A .

• Dimension

The dimension of a vector is the number of vectors in any basis.

This is well-defined → all bases for a space contain the same number of vectors

Dimensions of key subspaces of A (an $m \times n$ matrix of rank r)

- ↳ Column space $C(A)$: dimension r
- ↳ Nullspace $N(A)$: dimension $n - r$
- ↳ Row space : dimension r
- ↳ Zero space : dimension 0

→ The counting theorem : $\Rightarrow$ pivot variables + free variables = n
 $\Rightarrow$ i.e. $r + (n - r) = n$

• Beyond Column Vectors

The ideas of basis and dimension extend to any vector space, including matrix spaces and function spaces.

↳ The space of all 2×2 matrices

↳ has dimension 4 → require 4 independent matrices as a ^{basis} ~~basis~~

↳ The space of solutions to $y'' = -y$

↳ has dimension 2 → with basis = $\{\sin x; \cos x\}$

*** The generality of Basis and Dimension apply beyond $\mathbb{R}^n$.